Table 1. Detection performance under **synchronous** and **asynchronous** MAS execution across five attack scenarios. The LLM backbone is **gpt-oss-20b** for all baselines. Results are reported as **mean** ± **standard** deviation across runs. Best results are marked in **bold**.

Method	Synchronous MAS					Asynchronous MAS				
	F1↑	Prec.↑	Recall↑	FPR↓	AUROC↑	F1↑	Prec.↑	Recall↑	FPR↓	AUROC↑
CSQA **Stealthy Prompt Injection**										
TAM	0.28 ±0.11	0.24 ±0.10	0.33 ±0.12	0.61 ±0.09	41.67	0.19 ±0.09	0.16 ±0.08	0.24 ±0.10	0.72 ±0.08	18.43
PERM	0.38 ±0.12	0.33 ±0.11	0.44 ±0.13	0.49 ±0.10	51.56	0.27 ±0.10	0.23 ±0.09	0.32 ±0.11	0.58 ±0.09	36.82
G-Safeguard	0.74 ±0.33	0.72 ±0.36	0.82 ±0.35	0.20 ±0.33	94.32	0.46 ±0.31	0.42 ±0.33	0.55 ±0.32	0.45 ±0.30	70.58
BlindGuard	0.58 ±0.31	0.55 ±0.34	0.63 ±0.33	0.30 ±0.32	82.17	0.35 ±0.29	0.31 ±0.31	0.42 ±0.30	0.52 ±0.29	55.24
AcMAS	**0.95** ±0.03	**0.96** ±0.03	**0.95** ±0.03	**0.04** ±0.02	**99.41**	**0.94** ±0.03	**0.95** ±0.03	**0.94** ±0.03	**0.05** ±0.02	**98.76**
GSM8K **Stealthy Prompt Injection**										
TAM	0.27 ±0.10	0.23 ±0.09	0.32 ±0.11	0.63 ±0.09	45.14	0.18 ±0.08	0.15 ±0.08	0.23 ±0.09	0.74 ±0.08	17.62
PERM	0.37 ±0.12	0.32 ±0.11	0.43 ±0.12	0.51 ±0.10	50.23	0.26 ±0.10	0.22 ±0.09	0.31 ±0.10	0.60 ±0.09	35.47
G-Safeguard	0.71 ±0.31	0.69 ±0.34	0.80 ±0.33	0.23 ±0.31	92.48	0.44 ±0.30	0.40 ±0.32	0.52 ±0.31	0.47 ±0.30	66.73
BlindGuard	0.56 ±0.30	0.52 ±0.32	0.61 ±0.31	0.33 ±0.31	80.21	0.33 ±0.28	0.29 ±0.30	0.40 ±0.29	0.54 ±0.29	53.19
AcMAS	**0.95** ±0.03	**0.95** ±0.03	**0.95** ±0.03	**0.05** ±0.02	**99.36**	**0.94** ±0.03	**0.94** ±0.03	**0.94** ±0.03	**0.06** ±0.02	**98.52**
InjecAgent **Tool Manipulation Attack**										
TAM	0.51 ±0.28	0.47 ±0.26	0.57 ±0.29	0.38 ±0.27	50.20	0.34 ±0.25	0.30 ±0.23	0.40 ±0.26	0.54 ±0.26	35.78
PERM	0.64 ±0.29	0.60 ±0.27	0.69 ±0.30	0.26 ±0.28	84.60	0.43 ±0.27	0.39 ±0.25	0.49 ±0.28	0.45 ±0.27	58.34
G-Safeguard	0.69 ±0.30	0.66 ±0.32	0.78 ±0.31	0.26 ±0.30	90.64	0.41 ±0.29	0.37 ±0.31	0.49 ±0.30	0.50 ±0.29	61.42
BlindGuard	0.54 ±0.29	0.49 ±0.31	0.59 ±0.30	0.35 ±0.30	78.09	0.31 ±0.27	0.27 ±0.29	0.37 ±0.28	0.56 ±0.28	50.33
AcMAS	**0.93** ±0.04	**0.94** ±0.04	**0.93** ±0.04	**0.06** ±0.03	**98.71**	**0.92** ±0.04	**0.93** ±0.04	**0.92** ±0.04	**0.07** ±0.03	**97.58**
PoisonRAG **Memory Poisoning Attack**										
TAM	0.27 ±0.12	0.23 ±0.06	0.32 ±0.10	0.62 ±0.11	42.53	0.18 ±0.08	0.15 ±0.08	0.23 ±0.09	0.73 ±0.08	17.21
PERM	0.36 ±0.10	0.31 ±0.15	0.42 ±0.12	0.52 ±0.11	49.87	0.25 ±0.10	0.21 ±0.09	0.30 ±0.10	0.61 ±0.09	34.63
G-Safeguard	0.72 ±0.32	0.70 ±0.35	0.81 ±0.34	0.22 ±0.32	93.27	0.43 ±0.31	0.39 ±0.33	0.51 ±0.32	0.48 ±0.30	64.81
BlindGuard	0.57 ±0.30	0.53 ±0.33	0.62 ±0.32	0.32 ±0.31	81.14	0.33 ±0.29	0.29 ±0.31	0.39 ±0.30	0.55 ±0.29	52.06
AcMAS	**0.95** ±0.03	**0.96** ±0.03	**0.95** ±0.03	**0.04** ±0.02	**99.48**	**0.94** ±0.03	**0.95** ±0.03	**0.94** ±0.03	**0.05** ±0.02	**98.69**
HotPotQA **Memory Poisoning Attack**										
TAM	0.26 ±0.10	0.22 ±0.09	0.31 ±0.11	0.64 ±0.09	41.11	0.17 ±0.10	0.14 ±0.09	0.22 ±0.12	0.75 ±0.18	16.87
PERM	0.33 ±0.12	0.32 ±0.11	0.41 ±0.12	0.52 ±0.10	48.34	0.24 ±0.08	0.24 ±0.14	0.35 ±0.10	0.57 ±0.11	34.21
G-Safeguard	0.70 ±0.31	0.67 ±0.33	0.79 ±0.32	0.25 ±0.31	91.53	0.42 ±0.30	0.38 ±0.32	0.50 ±0.31	0.49 ±0.30	63.47
BlindGuard	0.55 ±0.30	0.50 ±0.32	0.60 ±0.31	0.34 ±0.31	79.62	0.32 ±0.28	0.28 ±0.30	0.38 ±0.29	0.56 ±0.29	51.28
AcMAS	**0.94** ±0.03	**0.95** ±0.03	**0.94** ±0.03	**0.05** ±0.02	**98.94**	**0.93** ±0.03	**0.94** ±0.03	**0.93** ±0.03	**0.06** ±0.02	**98.61**